

SamPave V2.0 Professional Baseline — User Manual

CRITICAL SAFETY DISCLAIMER

IMPORTANT:

SamPave is a professional decision-support engineering tool. All empirical design projections, thickness suggestions, material specifications, and mechanistic calculations generated by this software are for engineering screening and decision-support only. Final pavement designs, thickness profiles, and mix configurations must be independently verified, reviewed, and signed off by a licensed and qualified Professional Pavement Engineer before field execution, bidding, or construction.

1. Introduction & System Overview

SamPave Engineering Suite V2.0 Professional Baseline is a Qt-based desktop application designed to streamline the flexible pavement design workflow in accordance with the Indian Roads Congress (IRC) standards, specifically **IRC:37-2018** (Guidelines for the Design of Flexible Pavements) and Marshall mix design standards (**IRC:111-2009**).

The application integrates traffic analysis, subgrade estimation, catalog lookup, layer structure customization, stabilized cement-treated base/sub-base design, engineering compatibility checker, and the IITPAVE mechanistic analysis engine into a unified database-driven desktop suite.

2. Module Directory

2.1 Project Hub & Project Creation

- **Purpose:** The central workspace manager. Allows creation, duplication, revisioning, deletion, and management of highway design projects.
- **Required Inputs:**
 - Name of Work (Project Name)
 - Client Name (e.g., NHAI, State PWD)
 - Consultant Name
 - Submitted By (Lead Engineer)
 - Agency/Contractor
 - Mix Type Specification (e.g., DBM-I, DBM-II, BC-I, BC-II)
 - Binder Grade Selection (VG-30, VG-40, PMB-70)
- **Engineering Practice Source:** Comes from client tender files, contract documents, work orders, and consulting agreements.

- **Logic & Design Rules:**

- Project records are persisted in a local SQLite database (`pavement.db`).
- Upon project locking, the state is frozen (`locked = 1`), preventing accidental edits to any calculation module.
- Revisioning clones child records and increments the revision number, linking back to the parent project ID.

- **Step-by-Step Usage:**

1. Launch SamPave and open the Project Hub.
2. Click **Create New Project**.
3. Enter project metadata (Work Name, Client, Submitted By).
4. Select the desired Mix Type and Binder Grade from the dropdowns.
5. Click **Create Project**. The app opens the Project Dashboard.

- **Output Meaning:** Establishes a database record with a unique Project ID that binds all subsequent traffic, structural, stabilized, and mix design data.

- **Common Mistakes:**

- Entering duplicate project names, causing confusion during audit.
 - Failing to set the correct binder grade or mix type early, which affects default elastic modulus parameters.
 - **Validation Method:** Confirm the project appears in the Hub list and database `projects` table has correct metadata.
 - **Safety Disclaimer:** Project metadata is for reporting trace purposes only. Ensure project boundaries and client titles match contract records exactly.
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2.2 Traffic Analysis & MSA Calculation

- **Purpose:** Computes the cumulative design traffic in Million Standard Axles (MSA) over the design life of the pavement.

- **Required Inputs:**

- Initial Commercial Vehicles Per Day (CVPD) in both directions.
- Annual Growth Rate (\$r\$ in %).
- Design Life (\$n\$ in years).
- Vehicle Damage Factor (VDF).
- Lane Distribution Factor (LDF).

- **Engineering Practice Source:**

- CVPD comes from a 7-day, 24-hour classified traffic volume count survey (IRC:9).
- Growth rate comes from econometric data or historic traffic trends.
- VDF comes from axle-load survey statistics or default tables in IRC:37.

- LDF comes from lane count configurations defined in IRC:37 Clause 4.4.

- **Formula & Logic:**

$$N = \frac{365 * A * [(1+r)^n - 1]}{r} * D * F * 10^{-6}$$

Where:

- N = Cumulative design traffic (MSA)
- A = Initial CVPD after construction $= P * (1+r)^x$ (where P is traffic count and x is years between count and completion)
- r = Annual growth rate decimal
- n = Design life (years)
- D = Lane Distribution Factor (LDF)
- F = Vehicle Damage Factor (VDF)
- *Linear Guard:* If $r = 0$, the formula simplifies to:

$$N = 365 * A * n * D * F * 10^{-6}$$

- **Step-by-Step Usage:**

1. Navigate to the **Traffic Analysis** page.
2. Input the CVPD, annual growth rate, design life, VDF, and LDF.
3. Click **Compute Traffic**. The page displays the cumulative design traffic (MSA) and growth factor.
4. Click **Save & Continue**.

- **Output Meaning:** The computed design traffic (MSA) determines the structural catalog thickness requirements.

- **Common Mistakes:**

- Entering the growth rate as a decimal (e.g., entering 0.05 instead of 5.0 for 5%).
- Confusing LDF values (e.g., using 0.75 for a single-carriageway 2-lane road instead of 0.50).

- **Validation Method:** Hand-calculate using Eq. above and compare. Error must be 0.0000%.

- **Safety Disclaimer:** Traffic forecasts are subject to high variance. Sensitivity analysis for VDF and growth rate should be conducted.
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2.3 CBR & Subgrade Mr Calculation

- **Purpose:** Translates laboratory soaked California Bearing Ratio (CBR) values into the design resilient modulus (M_r) of the subgrade.

- **Required Inputs:**

- Soaked CBR (%) of the subgrade soil.

- **Engineering Practice Source:** Laboratory 4-day soaked CBR tests conducted on representative subgrade soil samples (IS 2720 Part 16).

- **Formula & Logic:**

- For $\text{CBR} \leq 5.0\%$:

$$M_r = 10 * \text{CBR} \text{ (in MPa)}$$

- For $\text{CBR} > 5.0\%$:

$$M_r = 17.6 * \text{CBR}^{0.64} \text{ (in MPa)}$$

- *Lower boundary guard:* CBR must be $> 0.0\%$. Values $\leq 0.0\%$ raise a **ValueError**.

- **Step-by-Step Usage:**

1. Navigate to the **Subgrade Properties** page.
2. Enter the design subgrade soaked CBR (%).
3. Click **Calculate Modulus**. The page displays the resilient modulus (M_r) in MPa.
4. Click **Save**.

- **Output Meaning:** Represents the elastic support capability (M_r) of the subgrade soil, which is the foundational support layer in the pavement system.

- **Common Mistakes:**

- Using unsoaked CBR values, which overestimate subgrade support under monsoon conditions.
- Entering soil classifications without laboratory CBR verification.

- **Validation Method:** Compare calculated M_r against empirical formulas.

- **Safety Disclaimer:** Subgrade modulus is highly moisture-sensitive. Ensure appropriate drainage design accompanies the selected modulus value.

2.4 IRC Catalogue Design

- **Purpose:** Automatically suggests a structural pavement composition based on the computed design traffic (MSA) and subgrade CBR.

- **Required Inputs:** None (automatically loaded from Traffic and Subgrade modules).

- **Engineering Practice Source:** The database represents digitized skeleton plates of **IRC:37-2018 Plates 1 to 4**.

- **Logic & Design Rules:**

- The catalog suggestions scale thicknesses logically: higher traffic and weaker subgrade soils trigger thicker pavement profiles.
- Granular layers (WMM and GSB) provide the sub-base/base support.
- Dense Bituminous Macadam (DBM) and Bituminous Concrete (BC) provide the bituminous surface.
- *Capping Warning:* Design traffic exceeding 150 MSA generates an engineering warning that the catalog limits are exceeded, and mechanistic validation is mandatory.

- **Step-by-Step Usage:**

1. Navigate to the **Catalogue Design** tab.
 2. The page reads the active design MSA and subgrade CBR.
 3. Click **Lookup Catalogue Composition**.
 4. The system displays the recommended thicknesses for BC, DBM, WMM, and GSB.
- **Output Meaning:** Provides the standard baseline thickness composition suggested by the IRC code plates.
 - **Common Mistakes:**
 - Adopting catalog suggestions directly for construction without running a mechanistic strain verification.
 - **Validation Method:** Cross-check suggestions against the tables in IRC:37.
 - **Safety Disclaimer:** The catalog provides indicative designs. Regional variations, local aggregate quality, and climate factors must be evaluated.
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2.5 Structural Pavement Design

- **Purpose:** Allows customization of the pavement layer thicknesses, material properties (moduli), and Poisson's ratios, and performs structural calculations.
- **Required Inputs:**
 - Number of layers (BC, DBM, WMM, GSB, Subgrade).
 - Thickness of each layer (mm).
 - Elastic Modulus of each layer (MPa).
 - Poisson's ratio of each layer.
- **Engineering Practice Source:** Moduli and Poisson's ratios are sourced from laboratory dynamic modulus tests, resonant column tests, or default values in IRC:37 Clause 6.2.
- **Logic & Design Rules:**
 - Allows manual overrides of catalogue values to adjust to site-specific conditions.
 - Validates physical thickness limits (e.g. thickness must be ≥ 0).
- **Step-by-Step Usage:**
 1. Navigate to the **Structural Pavement Design** page.
 2. Review the suggested catalogue thickness composition.
 3. To change a layer thickness or modulus, edit the value directly in the grid.
 4. Click **Compute Structural Design**.
- **Output Meaning:** The customized layer profile, which serves as the geometry and elastic input for the IITPAVE mechanistic simulation.
- **Common Mistakes:**
 - Inputting a lower layer with a significantly higher modulus than the upper layer without crack-relief design (stiffness inversion risk).
 - Entering unrealistic Poisson's ratios (e.g. 0.50, which causes mathematical singularities).

- **Validation Method:** Confirm the total thickness is the sum of individual layer thicknesses.
- **Safety Disclaimer:** Custom elastic moduli must represent actual pavement materials. Stiffening a layer on paper does not translate to field performance without strict quality control.

2.6 Bituminous & Marshall Mix Design

- **Purpose:** Analyzes laboratory Marshall compaction samples to determine volumetric properties (\$V_a\$, \$VMA\$, \$VFB\$) and the Optimum Binder Content (OBC).
- **Required Inputs:**
 - Specific gravity of coarse aggregates, fine aggregates, and bitumen.
 - Blend ratios of aggregates.
 - Bulk specific gravity (\$G_{mb}\$) of compacted mix specimens at varying bitumen contents (\$P_b\$).
 - Maximum theoretical specific gravity (\$G_{mm}\$) of loose specimens.
 - Stability and flow readings.
- **Engineering Practice Source:** Laboratory Marshall mix design tests (ASTM D6927 / IRC:111).
- **Formula & Logic:**
 - Air Voids (\$V_a\$):

$$V_a = \left(1 - \frac{G_{mb}}{G_{mm}}\right) \times 100$$

- Voids in Mineral Aggregate (\$VMA\$):

$$VMA = 100 - \frac{G_{mb}}{P_s} \times G_{sb}$$

(where \$P_s = 100 - P_b\$ is the aggregate percentage, and \$G_{sb}\$ is the combined aggregate bulk specific gravity)

- Voids Filled with Bitumen (\$VFB\$):

$$VFB = \frac{VMA - V_a}{VMA} \times 100$$

- **OBC Determination:** Cubic spline interpolation calculates the OBC, representing the binder content that satisfies the target air voids (typically 4.0% for DBM/BC).

- **Step-by-Step Usage:**

1. Navigate to the **Mix Design** module.
2. Input laboratory aggregate specific gravities and blend ratios.
3. Enter aggregate weight data, loose specimen flask weight details (for \$G_{mm}\$), and bulk specimen measurements (for \$G_{mb}\$).
4. Input stability (kN) and flow (mm) measurements.
5. Click **Compute Mix Design**. The system generates volumetric plots and highlights the OBC.

- **Output Meaning:** The mix recipe (OBC) and volumetric checks confirming compliance with IRC:111/MORTH specifications.
 - **Common Mistakes:**
 - Inputting chaotic specific gravities, causing cubic splines to warp or yielding unphysical volumetric parameters.
 - **Validation Method:** Check calculated values against equations. Error must be 0.0000%.
 - **Safety Disclaimer:** Lab mix design parameters must be verified through trial tracks and field density testing.
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2.7 CTB/CTS Stabilized Pavement Wizard

- **Purpose:** Suggests and validates cement-treated base (CTB) and cement-treated sub-base (CTS) layers, evaluating thickness savings and cracking risks.
- **Required Inputs:**
 - Cement-Treated Base thickness (mm) and modulus (MPa).
 - Unconfined Compressive Strength (UCS in MPa) of the CTB.
 - Cement-Treated Sub-base thickness (mm) and modulus (MPa).
 - Granular Sub-base (GSB) support layer thickness (mm).
- **Engineering Practice Source:** Sourced from laboratory cement-aggregate mix trials (UCS tests) and specifications in IRC:37 Section 9.
- **Logic & Design Rules:**
 - Compares the CTB/CTS stack against conventional flexible structures.
 - Computes nominal thickness savings (typically 30-50% reduction in bituminous cover thickness).
 - Flags a warning if UCS is zero (missing UCS).
 - Flags modular ratio and thin stabilized layer risks.
- **Step-by-Step Usage:**
 1. Navigate to the **Stabilized Design** tab.
 2. Configure CTB thickness, modulus, and UCS.
 3. Configure CTS thickness and modulus.
 4. Click **Calculate Stabilized Composition**.
 5. Review side-by-side comparison.
- **Output Meaning:** Provides structural suggestions and thickness reductions for stabilized pavements.
- **Common Mistakes:**
 - Using high CTB stiffness directly over a weak subgrade support, causing fatigue cracking of the stabilized slab.
- **Validation Method:** Compare composition thickness totals against design criteria.

- **Safety Disclaimer:** Cement-stabilized bases are highly prone to shrinkage cracking. Crack-relief interlayers (e.g. SAMI or aggregate layer) are critical.

2.8 Engineering Intelligence Checker

- **Purpose:** Screens pavement designs for structural layer compatibility, modular ratios, and stiffness anomalies before report finalization.
- **Required Inputs:** None (automatically evaluates active composition).
- **Engineering Practice Source:** Sourced from pavement mechanics principles and IRC:37 Section 6.
- **Logic & Design Rules:**
 - **Engineering Screening Score:** Starts at 100.0, and deducts points for design risks:
 - *Unusual thickness:* BC outside 25-60mm (-5), DBM outside 50-200mm (-5), CTB/CTS ≤ 100 mm (-15).
 - *Non-standard Poisson's ratio:* Stabilized layer Poisson outside 0.15-0.35 (-10).
 - *Bad modular ratio:* Base-to-subbase modulus ratio ≥ 4.0 (-15).
 - *Stiff layer over weak support:* CTB/CTS over support ≤ 500 MPa (-20).
 - *Stiffness inversion:* Lower layer stiffer than upper layer (-15).
 - *Bottom layer over subgrade:* Modulus ratio ≥ 4.0 (-15).
- **Step-by-Step Usage:**
 1. Click **Run Intelligence Check** or review the results on the **Dashboard**.
 2. The Checker displays the **Engineering Screening Score (0-100)** and flags compatibility warnings.
- **Output Meaning:** An indicator of the design's structural harmony. High score means low risk.
- **Common Mistakes:**
 - Treating the score as a final approval or pass/fail stamp. It is a screening tool.
- **Validation Method:** Trace deductions using the rules above.
- **Safety Disclaimer:** A 100/100 score does not guarantee structural performance. Mechanistic validation (IITPAVE) remains mandatory.

2.9 IITPAVE Integration Manager

- **Purpose:** Orchestrates, validates, and parses execution files of the Indian Roads Congress mechanistic design engine (**IITPAVE.exe**).
- **Required Inputs:**
 - **IITPAVE.exe** executable binary path.
 - Elastic properties of the layers (thickness, modulus, Poisson).
 - Wheel load configuration (default 20,000 N per wheel at 0.8 MPa contact pressure).
- **Engineering Practice Source:** The calculation kernel (**IITPAVE.exe**) is standard software distributed by the Indian Roads Congress.

- **Logic & Design Rules:**

- **Decision Support Mode:** Enabled if **IITPAVE.exe** is missing. IITPAVE run is blocked, and placeholder reports are generated.

- **Mechanistic Verified Mode:** Enabled only when a valid path to **IITPAVE.exe** is configured, execution completes successfully, and output parser successfully extracts strains.

- **Step-by-Step Usage:**

1. Navigate to the **IITPAVE Manager** page.
2. Configure the path to your local **IITPAVE.exe** binary.
3. Click **Validate Path**.
4. If valid, the system activates **Mechanistic Verified Mode** for all calculations.

- **Output Meaning:** Confirms execution compatibility and manages files (**iitp_inp.txt**, **iitp_out.txt**).

- **Common Mistakes:**

- Providing an invalid path or a corrupt file.
 - **Validation Method:** Check for version detection and path status on screen.
 - **Safety Disclaimer:** Always run on a licensed copy of **IITPAVE.exe**.
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2.10 DPR/Word Report Export

- **Purpose:** Generates a professional Detailed Project Report (DPR) containing all traffic, structural, stabilized, and mix design data.

- **Required Inputs:**

- Report ID
- Assumptions and limitations metadata
- Engineer sign-off checklist selections

- **Engineering Practice Source:** Sourced from standard DPR formats required by public works authorities.

- **Logic & Design Rules:**

- Exports directly to a formatted **.docx** file using native document generators.
- Forces inclusion of disclaimers, assumptions, limitations, and check lists.

- **Step-by-Step Usage:**

1. Click **Export DPR** on the project screen.
2. Review and select checklist items (e.g. "Traffic count verified").
3. Provide sign-off comments.
4. Choose the save path.
5. Click **Generate**.

- **Output Meaning:** A formatted Word document ready for internal review and consultant sign-off.

- **Common Mistakes:**

- Failing to review the generated document before submission.
 - **Validation Method:** Open the generated file and verify all tables are populated.
 - **Safety Disclaimer:** Generating a report does not imply professional approval. Sign-off spaces are manual and must be signed by hand.
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2.11 Submission Center & Project Locking

- **Purpose:** Locks project data to freeze design parameters and tracks consultancy review status.
 - **Required Inputs:**
 - Consultancy status selection (Draft, Under Review, Reviewed, Approved for Submission).
 - **Logic & Design Rules:**
 - Locking freezes calculations (**locked** = 1).
 - To edit a locked project, the user must click **Unlock** (requires confirming a warning) or create a new **Revision Copy** (incrementing revision number).
 - **Step-by-Step Usage:**
 1. Open the **Submission Center** tab.
 2. Select the desired **Review Status**.
 3. Click **Lock Project**.
 - **Output Meaning:** Restricts edits and secures calculation history.
 - **Common Mistakes:**
 - Unlocking a project and making silent edits without creating a new revision.
 - **Validation Method:** Try editing any input field on other pages; fields should be disabled.
 - **Safety Disclaimer:** "Approved for Submission" denotes internal consultancy workflow compliance only, not government or code certification.
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2.12 Submission ZIP Package

- **Purpose:** Packages the Word report, serialized JSON database inputs, raw IITPAVE files, and revision logs into a single package.
- **Required Inputs:** None.
- **Step-by-Step Usage:**
 1. Open the **Submission Center** page.
 2. Click **Export Package (.zip)**.
 3. Select the output directory.
 4. The system compiles the ZIP archive.

- **Output Meaning:** The delivery file containing both the human-readable report and machine-readable calculation history.
 - **Common Mistakes:**
 - Deleting the compiled Word report before archiving, causing a missing file error in the packager.
 - **Validation Method:** Extract the ZIP file and check for `README_Disclaimer.txt`, `ProjectInputs.json`, and the Word report.
 - **Safety Disclaimer:** The ZIP package is decision-support documentation. Final sign-off remains mandatory.
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3. Recommended Operational Settings

- **Dynamic modulus settings:** VG-30 bitumen binder dynamic modulus should typically range between 2000 and 3500 MPa at 35°C design temperature.
- **Granular layer ratios:** Modular ratio of base-to-subbase must be kept between 2.0 and 4.0 for structural compatibility.
- **Subgrade support:** Do not use CBR values below 2.0% without soil stabilization or subgrade borrow-soil replacement.